

Backend Development Schedule - UPDATED

Owner-first V1 | 12 weeks | 18 September - 10 December 2026

PROJECT BookingPartner.lk

DOCUMENT OWNER Fathima Suhana

START STATUS Day 1 completed; Day 2 pending as reported by you

PROJECT GOAL Owner onboarding to seat booking, verified payment, refund-to-source, and controlled soft launch.

What changed in this edition

Keeps the original twelve-week date range and week themes, but updates the daily tasks for your existing Next.js project, PostgreSQL schema, Prisma 7 setup, and your approved Owner V1 decisions. Replaces wallet refunds and Pusher with original-method refunds and seat-map polling.

This document is a work plan, not an executed database migration or proof of production readiness. The 10 December target is conditional on passing the launch gates.

YOUR VERY NEXT ACTION

Review package.json, prisma/schema.prisma, prisma7.config.ts, existing migrations and src/proxy.ts. Check the current Prisma client before creating auth or a singleton.

1 Current position and plan corrections

Use this page before ticking off any more daily tasks.

DAY 1 - DONE | Friday, 18 September 2026

You reported: PostgreSQL installed; bookingpartner_dev created; .env.local prepared; Prisma migration run; tables confirmed in pgAdmin. Retain the migration output and confirm the correct database was targeted.

DAY 2 - NEXT | Saturday, 19 September 2026

Prisma/package inspection -> DB client consistent with installed Prisma -> owner auth schema decision -> test DB call. Do not mark login or session complete until tested.

Important corrections

Create users, buses and booking tables again	Already have 143 application tables; review the real Prisma schema/migrations, do not duplicate.
lib/prisma.ts, lib/auth.ts everywhere	Server internals under src/backend/; Next.js routes under src/app/api/v1/.
Push seat changes with Pusher	V1 uses seat-map polling and conditional PostgreSQL seat acquisition.
Wallet credit on cancellation	No BookingPartner wallet. PayHere original-method refund where supported; recorded cash refund for counter bookings.
Create schedule-seat inventory	Use schedules -> dated trips -> trip_seats; keep existing layout and booking triggers.
Quick PayHere setup in Week 1	Initial provider review only; full verified payment work happens with booking state/ledger in Week 6.
Owner login without schema review	owners has no passwordHash in export: approve a credential/session design via Prisma migration first.

Source clarification: The attached schedule PDF contains 9 physical pages though its footers say 14. Its pages for Weeks 6 and 9-12 are not present. Their updated daily tasks below are reconstructed from the 84-day schedule you pasted in chat and your V1 decisions; earlier weeks retain the original themes.

2 The backend request flow and exact file paths

This is the structure that every daily API task should follow.

OWNER UI -> /api/v1/owner/buses/route.ts

-> BusController -> BusService -> BusRepository interface -> Prisma repository -> PostgreSQL -> safe API response -> frontend

Authentication, owner status, membership permission and ownership checks happen server-side. Frontend controls alone are not access control.

Recommended directory blueprint

```
D:\booking_partner\  
|-- prisma\schema.prisma  
|-- prisma\migrations\  
|-- src\app\owner\           (existing owner frontend)  
|-- src\app\owner-auth\      (existing owner auth UI)  
|-- src\app\api\v1\owner-auth\... (thin HTTP routes)  
|-- src\app\api\v1\owner\...   (thin HTTP routes)  
|-- src\backend\composition\  
|-- src\backend\core\errors\  
|-- src\backend\core\http\  
|-- src\backend\core\security\  
|-- src\backend\core\config\  
|-- src\backend\modules\auth\  
|-- src\backend\modules\owner\  
|-- src\backend\modules\bus\  
|-- src\backend\modules\schedule\  
|-- src\backend\modules\trip\  
|-- src\backend\modules\seat\  
|-- src\backend\modules\booking\  
|-- src\backend\modules\payment\  
|-- src\backend\modules\ledger\  
|-- src\backend\infrastructure\database\  
|-- src\backend\jobs\  
|-- src\proxy.ts             (already exists)  
|-- tests\  

```

Prisma / Auth.js note

Do not create src/app/api/auth/[...nextauth]/route.ts until you confirm Auth.js/NextAuth is installed and choose its integration. Do not copy an old Prisma singleton into Prisma 7 without inspecting the generator import and database adapter configuration.

3 Database changes - gated, not automatic

Existing 143-table PostgreSQL export is the baseline. Make reviewed Prisma migrations only when required by a specific feature.

Before owner login	Owner credential/password hash; secure revocable session; phone OTP challenge; reset tokens; verified phone timestamp; email/phone normalization after duplicate audit.
Before management invite	Reuse management_members and management_partner_connections; membership-scoped permission migration; single-use activation; no plaintext password by SMS/email.
Before bus create	Normalized registration plate unique among non-deleted buses; choose whether soft-delete exists, since current buses has no deletedAt.
Before online bookings	Seat-hold extension once (10 to 15 minutes total); coherent expiry of pending_payment; idempotency keys; snapshot cancellation terms.
Before PayHere	payment_attempts + payment_webhook_events with deduplication; merchant sandbox refund / partial refund support verification.
Before live money	ledger_accounts + ledger_transactions + append-only ledger_entries; settlement references; reconciliation discrepancy records.
Before counter booking	Decide whether LKR 100 is INCLUDED in total or ADDED to fare. Existing DB requires bookings.totalAmount = sum(booking_seats.price). Do not fake seat price.

DO NOT RUN THE STANDALONE 143-TABLE SQL SCRIPT AGAINST YOUR POPULATED DATABASE.

It is an export, not a Prisma migration-history replacement. Back up, inspect data, create reviewed Prisma migrations, test in bookingpartner_dev, then apply through the approved deployment workflow.

WEEK 01 | Setup and foundation

Fri, 18 Sep - Thu, 24 Sep 2026 | Local project configuration, safe Prisma infrastructure, Owner credential design, reliable error handling.

STATUS: Day 1 completed (user-reported). Day 2 and later are pending until tested.

Day 01 - Fri, 18 Sep [DONE] | Confirm database baseline

Record Day 1 as done; confirm the intended bookingpartner_dev connection, Prisma migration status, and localhost UI.

Done when: No migration drift or unexpected data loss.

Day 02 - Sat, 19 Sep [NEXT] | Inspect and connect Prisma

Read package.json, schema.prisma, prisma7.config.ts, existing API routes and proxy.ts. Implement src/backend/infrastructure/database/prisma.client.ts only as required by actual Prisma 7 setup.

Done when: One server-side API can query development PostgreSQL safely.

Day 03 - Sun, 20 Sep | Design owner authentication

Choose credential/session schema for owner email + password; identify needed phone OTP and reset records. Prepare migration proposal and authentication error codes.

Done when: Schema change checklist reviewed; no arbitrary auth tables added.

Day 04 - Mon, 21 Sep | Build owner auth vertical slice

Implement thin owner-auth login/me/logout routes through auth service/repository; use approved hashed credentials, secure cookies or chosen Auth.js session model.

Done when: Good/wrong credentials and logout work for local test owner.

Day 05 - Tue, 22 Sep | Phone OTP and owner status

Implement development OTP challenge flow if provider is ready; rate-limit and expire codes. Apply pending/review/active/suspended policies.

Done when: OTP reuse rejected; suspended owner cannot start new sales.

Day 06 - Wed, 23 Sep | Core API consistency

Create response wrapper, AppError/handler, Zod validation, request ID and secret-safe logs; keep src/proxy.ts aligned with Next.js version.

Done when: 400/401/403/409/500 are handled consistently.

Day 07 - Thu, 24 Sep | First-week smoke tests

Run login/session/logout/status tests; review existing bus routes and owner ID sourcing; document blocker list.

Done when: No critical broken authentication or configuration flow.

WEEK FINISH CHECK

Local DB and authenticated Owner vertical slice; no broken critical flow.

WEEK 02 | Authentication and users

Fri, 25 Sep - Thu, 01 Oct 2026 | Finish the full Owner signup path, customer auth basics, controlled management invitations and recovery.

Day 08 - Fri, 25 Sep | Owner onboarding API

Create pending owner with email/password and normalized phone; do not allow public request to set approval or verification flags.

Done when: Duplicate identity and unsafe role/status fields rejected.

Day 09 - Sat, 26 Sep | Owner phone verification

Implement send/verify once-at-signup OTP with attempts, expiry, single-use storage; define phone change re-verification.

Done when: Verified phone attached to correct pending owner.

Day 10 - Sun, 27 Sep | Verification documents

Allow pending owner profile and protected document upload; under-review responds to admin requests only.

Done when: Owner cannot view or modify another owner documents.

Day 11 - Mon, 28 Sep | Customer registration baseline

Use existing customers.passwordHash; validate and hash; define separate customer sessions/verification only where required.

Done when: New local customer signs in without owner access.

Day 12 - Tue, 29 Sep | Invite primary management user

Reuse member-owner connection; create single-use invitation and password setup, not emailed temporary plaintext password.

Done when: Owner cannot invite into another operator.

Day 13 - Wed, 30 Sep | Sub-user invites and permissions

Primary may invite sub only with permission; membership-scoped schedules/bookings/manual-booking flags, no financial access.

Done when: Wrong membership and expired invite denied.

Day 14 - Thu, 01 Oct | Account recovery test day

One-time reset tokens, session invalidation, owner approval tests and verification email/SMS failure handling.

Done when: Auth flows pass with local test data and safe errors.

WEEK FINISH CHECK

Owner and customer auth, phone OTP, management invitation and access gates tested.

WEEK 03 | Cities, routes, buses and layouts

Fri, 02 Oct - Thu, 08 Oct 2026 | Reuse existing bus and route records; Owner fleet operations enforce ownership and layout history.

Day 15 - Fri, 02 Oct | Cities and routes reads

Read existing cities, routes, route_stops and boarding_points, exposing only active public options.

Done when: Public dropdowns show real database records.

Day 16 - Sat, 03 Oct | Dated trip search baseline

Validate origin, destination, departure date; join schedules -> trips and return only bookable trips.

Done when: Search is based on the requested departure date.

Day 17 - Sun, 04 Oct | Owner bus list and create

Refactor existing /api/v1/owner/buses; derive ownerId from session and status gate; validate bus fields.

Done when: Owner A cannot list or create under owner B.

Day 18 - Mon, 05 Oct | Bus detail/edit

Preserve existing [id] route; check busId and ownerId; admin-only transfer process.

Done when: Changing URL bus ID does not disclose other owner data.

Day 19 - Tue, 06 Oct | Bus compliance and photos

Validate protected document upload, permissions and operator visibility; keep records in existing tables.

Done when: Unsupported files and cross-owner updates rejected.

Day 20 - Wed, 07 Oct | Seat layout versioning

Use bus_layouts and bus_layout_slots, preserve used-layout immutability; create new layout version when necessary.

Done when: Reopen layouts without changing historical trip seats.

Day 21 - Thu, 08 Oct | Bus workflow test

Test normalized duplicate plate rule if migration approved, approval flags, active layout and mixed owner accounts.

Done when: Owner CRUD/layout/ownership and constraint tests pass.

WEEK FINISH CHECK

Owner fleet CRUD and layout versions work against existing PostgreSQL.

WEEK 04 | Schedules and dated trips

Fri, 09 Oct - Thu, 15 Oct 2026 | Owners create schedule definitions, dated trips and their own trip_seats, then manage operational changes.

Day 22 - Fri, 09 Oct | Create schedule and trip

POST /api/v1/owner/schedules; check active owner, approved bus, route/time/fare; generate dated trip and trip_seats from layout.

Done when: Each dated trip has correct sellable seat inventory.

Day 23 - Sat, 10 Oct | List/filter/detail

Return owner-only schedules/trips, route/date/status filters; availability counted from trip_seats.

Done when: No cross-operator rows; count matches PostgreSQL.

Day 24 - Sun, 11 Oct | Repeat trips

Create dated trips for daily/weekly repeat request; use schedule_repeat_groups as defined, skip duplicates in one transaction.

Done when: Correct dates and unique trip seat inventories.

Day 25 - Mon, 12 Oct | Schedule conflict detection

Detect same bus overlapping departures on a date, including edits; block per approved rule.

Done when: No time clashes silently committed.

Day 26 - Tue, 13 Oct | Update time and crew

Authorize future trip changes; verify assigned crew and bus belong to operator; record trip_status_changes.

Done when: Affected passengers enumerated for SMS.

Day 27 - Wed, 14 Oct | Cancel trip safely

Stop sale, apply owner/admin case rule for existing bookings; start PayHere original-method refund or recorded operator cash refund. No wallet.

Done when: Booked and unbooked cases leave valid final states.

Day 28 - Thu, 15 Oct | Passenger manifest PDF

Owner-only per-trip PDF with permitted passenger/seat/boarding fields; omit unnecessary PII.

Done when: Unauthorized owner denied; zero and many passengers render.

WEEK FINISH CHECK

Owner trip lifecycle and refund initiation work without wallet assumptions.

WEEK 05 | Seat holding and polling

Fri, 16 Oct - Thu, 22 Oct 2026 | 10-minute hold, one extension to 15-minute total, atomic acquisition, coherent expiration and two-user tests.

Day 29 - Fri, 16 Oct | Atomic seat hold

POST /api/v1/seats/hold checks trip and seat; DB conditional update wins for one holder only.

Done when: Concurrent clients cannot both acquire one seat.

Day 30 - Sat, 17 Oct | Hold extension once

After PayHere redirect step, allow a single extension up to 15 minutes total; record extension usage in approved schema.

Done when: Second extension and invalid holder denied.

Day 31 - Sun, 18 Oct | Release hold

Release only the holder or authorized cleanup; confirmed bookings are never released by hold endpoint.

Done when: Repeat release is safe.

Day 32 - Mon, 19 Oct | Expiry cleanup

Cron handles expiresAt <= database now; expire related pending_payment record coherently before seat becomes sellable.

Done when: No active pending booking blocks reclaimed seat.

Day 33 - Tue, 20 Oct | Seat-map read API

GET /api/v1/trips/{tripId}/seats; return current safe status and treat expired hold accurately.

Done when: Opening map returns correct authoritative state.

Day 34 - Wed, 21 Oct | Polling UI integration

Poll while seat map visible; slow/stop in background; refetch after local hold/release; no Pusher.

Done when: Two tabs converge without refresh.

Day 35 - Thu, 22 Oct | Race and recovery tests

Test simultaneous holds, expiry, payment redirect, disconnected tab, cron rerun and late callbacks.

Done when: Only one valid active claimant per trip seat.

WEEK FINISH CHECK

One seat claimant, correct expiration, polling UI and race tests.

WEEK 06 | Payments and booking confirmation

Fri, 23 Oct - Thu, 29 Oct 2026 | PayHere sandbox, verified idempotent callbacks, cash distinction, original-method refunds and ledger.

Day 36 - Fri, 23 Oct | Pending booking endpoint

POST /api/v1/bookings with frozen price/policy and owner/actor provenance; request idempotency key.

Done when: One request cannot create duplicate pending bookings.

Day 37 - Sat, 24 Oct | Payment attempt and gateway

Record gateway payment attempt with booking, amount, currency, status; generate PayHere request server-side.

Done when: Secret never reaches browser.

Day 38 - Sun, 25 Oct | Verified webhook

Verify PayHere server callback, amount, currency, booking and provider reference; store deduped webhook event.

Done when: Invalid, mismatched and duplicate callbacks safe.

Day 39 - Mon, 26 Oct | Confirmation and ledger

In one DB transaction confirm booking/seat, record balanced ledger entries and payout liability; notify after commit.

Done when: One financial posting per successful attempt.

Day 40 - Tue, 27 Oct | Late-payment and reconciliation

If hold expired and seat reallocated, record exception and initiate verified refund-to-source; compare gateway records daily.

Done when: No late paid booking steals another seat.

Day 41 - Wed, 28 Oct | Booking details and cancellation

Customer-only list/detail; snapshot policy for eligibility; admin handles PayHere refund; counter cash refund recorded.

Done when: No wallet credit; actual method and state visible.

Day 42 - Thu, 29 Oct | Ticket and payment tests

PDF/QR ticket authorized; test success/failure/duplicate/partial refund feasibility in merchant sandbox.

Done when: No critical mismatch among ticket, seat, payment, ledger.

WEEK FINISH CHECK

PayHere-to-booking-to-ledger confirmation and refund exception controls.

WEEK 07 | Owner earnings, management and settings

Fri, 30 Oct - Thu, 05 Nov 2026 | Owner can safely view money, manage staff, update company/bank details and create cash counter bookings.

Day 43 - Fri, 30 Oct | Owner earnings summary

GET /api/v1/owner/earnings uses paid eligible bookings and ledger/settlement states by trip.

Done when: Amounts match audited fixtures.

Day 44 - Sat, 31 Oct | Payout breakdown

GET /api/v1/owner/payouts; group pending, held, paid with ownerId restrictions.

Done when: Owner A never views Owner B payments.

Day 45 - Sun, 01 Nov | Management list and invite UI

Complete owner/primary invitation screens and single-use activation already designed in Week 2.

Done when: Member only sees own operator membership.

Day 46 - Mon, 02 Nov | Permissions and assignment

Enforce operator-scoped flags; deactivate member and revoke sessions; crew-to-bus same-owner checks.

Done when: Wrong membership cannot create manual booking.

Day 47 - Tue, 03 Nov | Company and bank settings

Update profile; bank-change requests require re-auth/admin approval; mask bank details.

Done when: Previous active bank history preserved.

Day 48 - Wed, 04 Nov | Policy and notifications

Owner edits allowed cancellation and alert settings; snapshots of past bookings remain unchanged.

Done when: Existing booking refund terms not overwritten.

Day 49 - Thu, 05 Nov | Manual counter booking

Authorized owner/manager creates cash booking; LKR 100 platform fee per approved breakdown; no fake PayHere attempt.

Done when: Cash receivable and seat ledger entry match booking total.

WEEK FINISH CHECK

Owner finance, management permissions and counter booking are auditable.

WEEK 08 | Admin owner and financial APIs

Fri, 06 Nov - Thu, 12 Nov 2026 | Admin approval, verification, payouts, reports and audit without bypassing owner status rules.

Day 50 - Fri, 06 Nov | Owner review queue

Paginated /api/v1/admin/owners with pending, review, active, suspended and rejected states.

Done when: Admin authentication/permission required.

Day 51 - Sat, 07 Nov | Owner approval actions

Approve/reject/suspend and record audit; explicitly decide existing-booking action per affected trip.

Done when: Suspension never silently cancels unrelated travel.

Day 52 - Sun, 08 Nov | Bus verification

Review compliance docs/layout; approve or return for correction; only eligible buses show in search.

Done when: Owner cannot approve own bus.

Day 53 - Mon, 09 Nov | Financial release flow

List settlement batches, payout references and ledger evidence; approve/withhold with reason.

Done when: No double payout on retry.

Day 54 - Tue, 10 Nov | Refund and disputes queue

Review PayHere-to-source and cash operator refunds with proofs; audit each decision.

Done when: Operator/customer get only permitted refund status.

Day 55 - Wed, 11 Nov | Reports and support

List/assign/reply reports and owner verification queries, preserving history.

Done when: Staff cannot read unrelated private material.

Day 56 - Thu, 12 Nov | Admin analytics test

Bookings/day, routes, reconciled revenue and held liability; date filtering and pagination.

Done when: Numbers match sample financial journal.

WEEK FINISH CHECK

Admin actions audited; payout/refund queues permissioned.

WEEK 09 | Management, transfers, reviews, notifications

Fri, 13 Nov - Thu, 19 Nov 2026 | Permissioned management operations and customer-facing follow-up features.

Day 57 - Fri, 13 Nov | Management schedule reads

Primary/sub-user can view only membership-linked operator schedules and permitted routes.

Done when: Role alone never grants cross-owner access.

Day 58 - Sat, 14 Nov | Management manual booking

Allow with explicit manual_booking flag; reuse atomic seat and cash accounting path.

Done when: Cannot skip fee, authorizations or ledger.

Day 59 - Sun, 15 Nov | Transfer initiation

Define transfer code/token life cycle, ownership and confirmed-booking eligibility.

Done when: Token is one-time and short-lived.

Day 60 - Mon, 16 Nov | Transfer accept/expiry

Accept/decline safely and expire after policy-defined period; record provenance and notifications.

Done when: Double acceptance and stale token denied.

Day 61 - Tue, 17 Nov | Post-trip review create

Allow eligible passengers after travel, validate rating and owner reply boundaries.

Done when: No arbitrary review of untraveled trip.

Day 62 - Wed, 18 Nov | Review reads and rating

Bus/owner rating from permitted reviews; return public fields only.

Done when: Moderation state respected.

Day 63 - Thu, 19 Nov | Notification verification

SMS OTP/booking/refund/trip changes; email receipts/approval/payout; polling for seat map.

Done when: Retries do not generate duplicate sends.

WEEK FINISH CHECK

Management and notification flows enforce operator membership.

WEEK 10 | Security, concurrency and end-to-end testing

Fri, 20 Nov - Thu, 26 Nov 2026 | Fix defects before deployment; frontends must run on real APIs rather than old mocks.

Day 64 - Fri, 20 Nov | API checklist

List routes and test success, 400, 401, 403, 404, 409 and 500; request IDs.

Done when: Failures never leak credentials or stack traces.

Day 65 - Sat, 21 Nov | Ownership/IDOR tests

Owner, manager, admin, customer and cross-operator access to bookings, buses and bank data.

Done when: Every wrong actor denied.

Day 66 - Sun, 22 Nov | PayHere sandbox cases

Verify valid, failed, duplicate, late, mismatched amount, refund and unsupported partial refund case.

Done when: No duplicate journal posting.

Day 67 - Mon, 23 Nov | Seat/load tests

Race acquisition, cron replay, lazy expiry, time change and cancellation.

Done when: One final valid trip seat state.

Day 68 - Tue, 24 Nov | Frontend owner integration

Remove owner mock data screen by screen; test real loading, empty and error states.

Done when: Owner UI uses actual API contracts.

Day 69 - Wed, 25 Nov | Customer/admin integration

Connect booking confirmation, payouts, admin approval, refund status and notifications.

Done when: One shared source of truth.

Day 70 - Thu, 26 Nov | Full journey / go-no-go

Run owner signup -> approval -> bus -> trip -> search -> hold -> pay -> ticket -> refund.

Done when: All critical checks pass or launch blocked.

WEEK FINISH CHECK

Critical system end-to-end and concurrency tests pass.

WEEK 11 | Production readiness and deployment

Fri, 27 Nov - Thu, 03 Dec 2026 | Staged deployment and a separate production PostgreSQL database, with rollback rehearsals.

Day 71 - Fri, 27 Nov | Repo and CI

Protect main, add tests/build pipeline and deployment review; do not commit secret files.

Done when: Clean CI and reproducible build.

Day 72 - Sat, 28 Nov | Production DB provision

Create isolated managed PostgreSQL, backups and restricted credentials.

Done when: No dev/prod data mixing.

Day 73 - Sun, 29 Nov | Migration deployment rehearsal

Restore sample/staging backup and run committed migrations using deploy workflow.

Done when: Constraints/functions/triggers retained.

Day 74 - Mon, 30 Nov | Production secrets

Configure hosting environment with DB, session, OTP, SMS/email, PayHere and cron keys.

Done when: Secrets only on server.

Day 75 - Tue, 01 Dec | Provider callbacks and jobs

Set correct production webhook URL, verify callback, polling, refund and reconciliation jobs.

Done when: Duplicate deliveries safe.

Day 76 - Wed, 02 Dec | Staging security test

Test actual HTTPS, owner states, cron authorization and payment flow from Sri Lanka mobile network.

Done when: No critical public-data exposure.

Day 77 - Thu, 03 Dec | Operational runbook

Practice restore, reconciliation incident, delayed callback, operator suspension and rollback.

Done when: Named response and recovery procedure.

WEEK FINISH CHECK

Staging and production infrastructure are separated and recoverable.

WEEK 12 | Operator onboarding and conditional soft launch

Fri, 04 Dec - Thu, 10 Dec 2026 | Gradual operator onboarding, supervised transaction and financial sign-off.

Day 78 - Fri, 04 Dec | First operator preparation

Prepare company, phone verification, owner approval, approved fleet, routes and valid dated trips.

Done when: All artifacts match production DB.

Day 79 - Sat, 05 Dec | Operator walkthrough

Supervised owner/management actions, bus edit, manual booking, payout view and access checks.

Done when: Only own data visible.

Day 80 - Sun, 06 Dec | Controlled real-money pilot

With permission and oversight, verify PayHere payment, booking, seat, ledger, ticket and SMS.

Done when: Real money fully reconciled.

Day 81 - Mon, 07 Dec | Original-method refund pilot

Test supported online refund and recorded counter refund with agreed policy; validate owner/admin visibility.

Done when: Refund outcome equals journal.

Day 82 - Tue, 08 Dec | Incident correction

Resolve production-only bugs; run critical flow again after fixes.

Done when: No unresolved critical issue.

Day 83 - Wed, 09 Dec | Small operator cohort

Onboard 2-3 approved operators; verify independent permissions and seat states.

Done when: No cross-operator leakage.

Day 84 - Thu, 10 Dec | Launch readiness gate

Review backups, owner rules, seat races, payments, refunds, reconciliation, security and operator support.

Done when: Launch only if gates pass; date may move.

WEEK FINISH CHECK

Conditional launch sign-off after real-money reconciliation.

Appendix A Owner module - first API vertical slice

The file paths are proposed and should be reconciled with the actual repository before creating new files.

Frontend	src/app/owner/ + src/app/owner-auth/
API route	src/app/api/v1/owner/buses/route.ts
Controller	src/backend/modules/bus/bus.controller.ts
Service	src/backend/modules/bus/bus.service.ts
Repository contract	src/backend/modules/bus/bus.repository.interface.ts
Prisma repository	src/backend/infrastructure/database/repositories/bus.prisma.repository.ts
Client	src/backend/infrastructure/database/prisma.client.ts
Wiring	src/backend/composition/container.ts
Owner policy	src/backend/modules/owner/owner.policy.ts
Shared HTTP/errors	src/backend/core/http/ and src/backend/core/errors/
Test	tests/integration/owner/ + tests/security/ownership-access.test.ts

Practical test owner setup

Create one active test owner and one pending or suspended test owner after credential/schema work is approved. Then create a second operator to test isolation. Never use a plaintext password in a fixture or store production secrets in test files.

Appendix B Before-you-build checklist

For Day 2 and the first approved migration.

Project and DB inventory

- ☐ package.json Prisma/Auth.js versions
- ☐ prisma/schema.prisma client generator and models
- ☐ prisma7.config.ts and adapter details
- ☐ committed migrations, especially trigger/check SQL
- ☐ current src/proxy.ts and existing owner routes

Owner authentication decisions

- ☐ owner password hash storage approach
- ☐ phone OTP purpose, expiry and retry policy
- ☐ session store and revocation
- ☐ owner status access matrix
- ☐ password recovery and phone-change verification

Owner scope and operator management

- ☐ reuse management_partner_connections
- ☐ membership-scoped permissions migration
- ☐ one-time invites; never SMS reusable plaintext password
- ☐ cross-owner bus/crew/booking tests

Booking and money gates

- ☐ LKR 100 fee included vs added to fare
- ☐ paid vs manual booking provenance
- ☐ 10 -> 15 minute hold extension
- ☐ refund/partial refund merchant capability
- ☐ frozen cancellation policy and balanced ledger

Appendix C V1 business rules and final launch gates

Features marked V2 must not accidentally enter V1 through old tasks or stale PDF examples.

Owner sign-in	Email + password; signup phone OTP; reverify phone change.
Owner status	Pending limited; under_review reply/read; active operational; suspended read-only + admin decision about existing trips.
Management	Owner invites primary, primary invites sub with permission; one operator per member in V1, membership-ready design.
Manual booking	Counter cash, LKR 100 platform fee; no fake gateway payment; no agent commission.
Online payment	PayHere enabled merchant methods; no app-managed wallet; no bank transfer checkout.
Refund	Original online method where supported + admin action; operator returns counter cash with proof.
Seats	10 minutes, one extension to 15 total, DB clock, coherent booking expiry, polling map.
Finance	Balanced append-only financial ledger, auditable settlement, daily PayHere reconciliation.
Commissions	Agent commission stays disabled in V1; enable in V2 after validated accounting cycles.
Messaging	SMS OTP/booking/refund/trip; email receipts/approval/payout; no Pusher requirement.

GO / NO-GO: Do not launch merely because the calendar says 10 December.

Require signed-off seat concurrency, verified callback/idempotency, late-payment refund, original-method refund policy, balanced ledger, backup/restore, operator isolation, audit logs, production secrets and operator support.

Appendix D Source and scope notes

Distinguish material supplied by you from new technical recommendations.

Document basis

1. Attached BookingPartner.lk Backend Development Schedule PDF: 9 physical pages (page labels indicate 14), covering the general structure, Weeks 1-5, 7 and 8.
2. Your pasted 84-day / 12-week task list in chat supplied the gaps and exact overall date span; its older wallet and Pusher tasks are superseded by your later V1 decisions.
3. Uploaded BookingPartner-143-tables-postgresql.sql: existing tables, checks, indexes, booking triggers, owner roles and management linkage.
4. Owner V1 decisions supplied in chat: OTP + email/password; invitation-only management; owner status, polling, counter cash, PayHere, refund-to-source, ledger and no BookingPartner wallet.

Important technical distinctions

The SQL export is not the current Prisma schema or proof that the app runs. Prior to code changes compare live development database, Prisma schema, migration history, deployed behavior and actual API source. Table additions in this plan are recommendations, not already-applied migrations.

No owner payout account or refund merchant capability is assumed to be active until independently tested. Every money-changing task has an audit, failure and reconciliation test.

WHAT TO DO FIRST

Complete Day 2 as a configuration review + one working DB query. Once credentials and session schema are approved, implement owner login as one tested request-to-repository vertical slice.